Gain-vs-charge correction: effect on the Am-241 lines

Data

AMTEst, 10 images (Image 8, 500 skips, board 106; EXT1, EXT3, EXT4 - EXT2 is flooded). Cluster files from run_WADERs (WADERs_analysis_fz_files/root), corrected with gain_vs_charge.py (apply). All clusters.

What the correction does

The gain falls with the charge in a pixel (here 2.5-3.6% lower at 1000 e-). It is measured from the images themselves (spacing of the electron comb at high charge) and used to convert every cluster pixel from ADU to electrons, instead of one gain from the 0e-/1e- peaks. One curve per CCD for all 10 images, so image-to-image differences of the fitted 0e-/1e- gain disappear too.

Gain curves measured

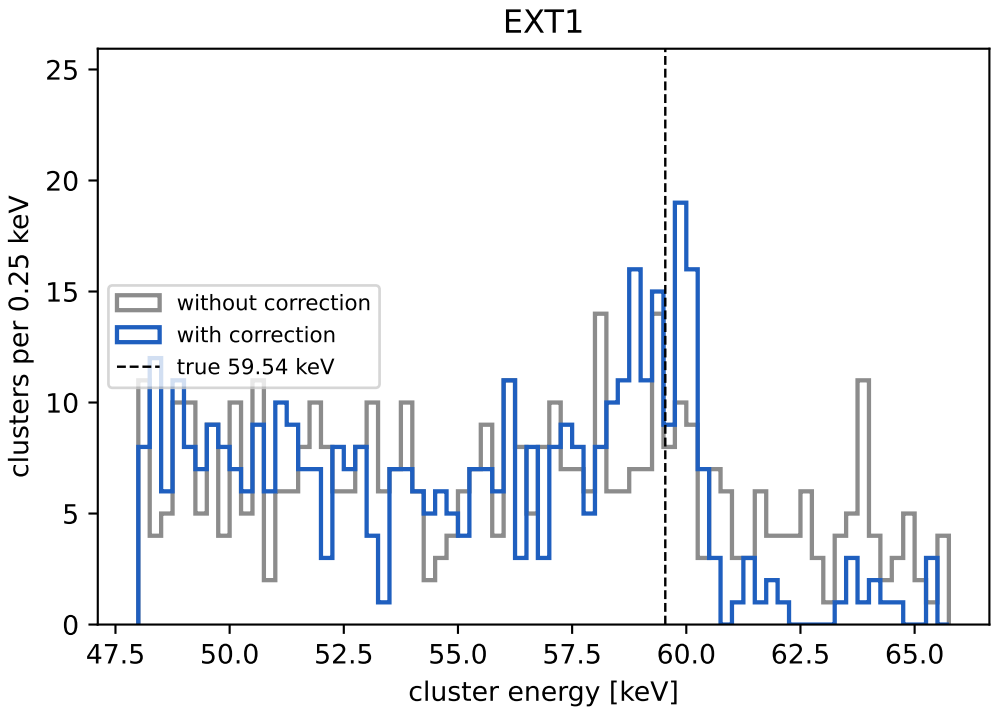
EXT1: 3.6% lower at 1000 e-, low-charge gain 1.371. EXT3: 2.5% lower at 1000 e-, low-charge gain 1.303. EXT4: 3.4% lower at 1000 e-, low-charge gain 1.312. EXT1: the 0e-/1e- gains panaSKImg fitted per image scatter from 1.13 to 1.32 (very low dark current, weak 1e- peak), so its energies were too HIGH and come down; on EXT3/EXT4 they were fine and the correction mainly undoes the high-charge gain drop, so energies go UP.

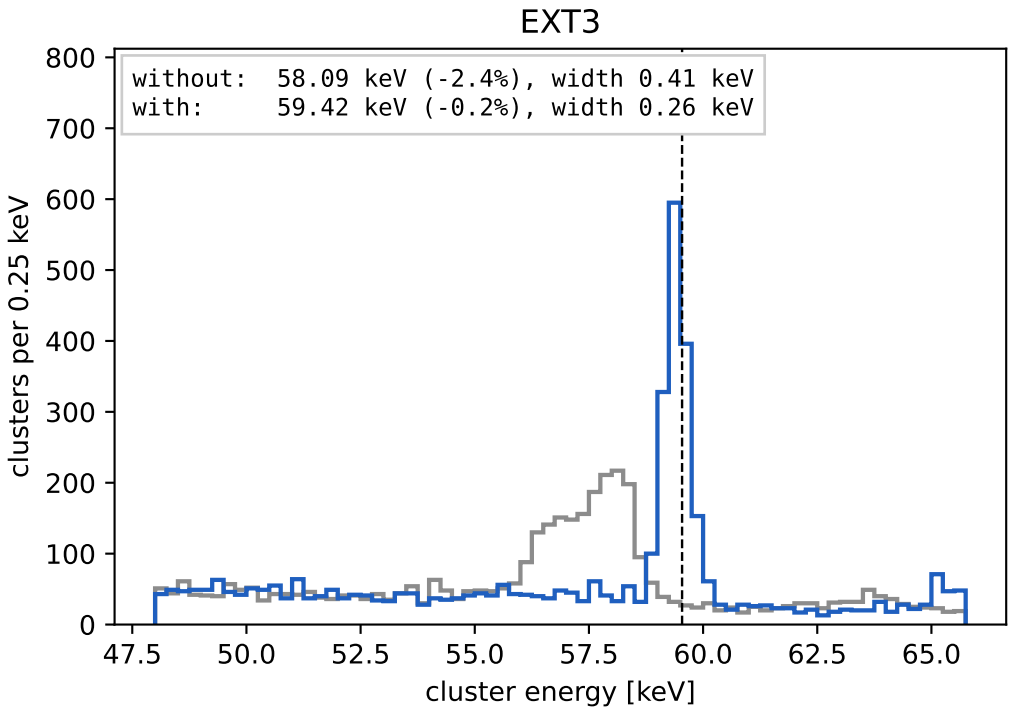
Fitted line positions, measured / true energy

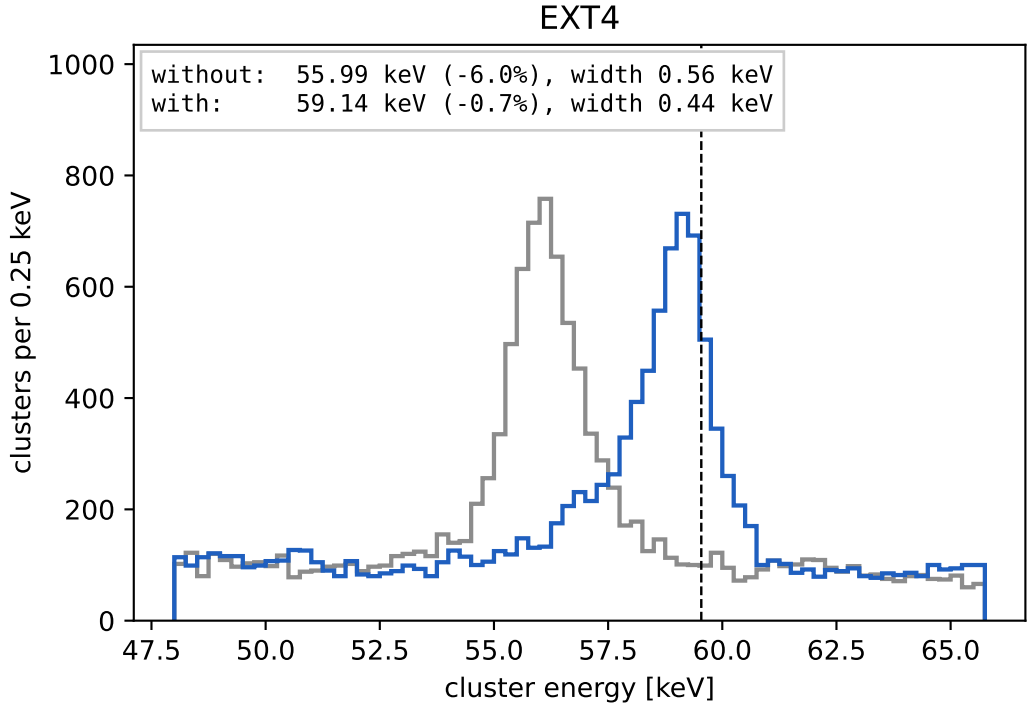
line	true keV	EXT1 without	with	EXT3 without	with	EXT4 without	with
Np Lalpha	13.93	-	-	-	-	-	-
Np Lbeta	17.75	-	-	-	-	0.942	0.985
Np Lgamma	20.80	-	-	-	-	-	-
gamma 26.3	26.34	-	-	-	-	0.940	0.990
gamma 59.5	59.54	-	-	0.976	0.998	0.940	0.993

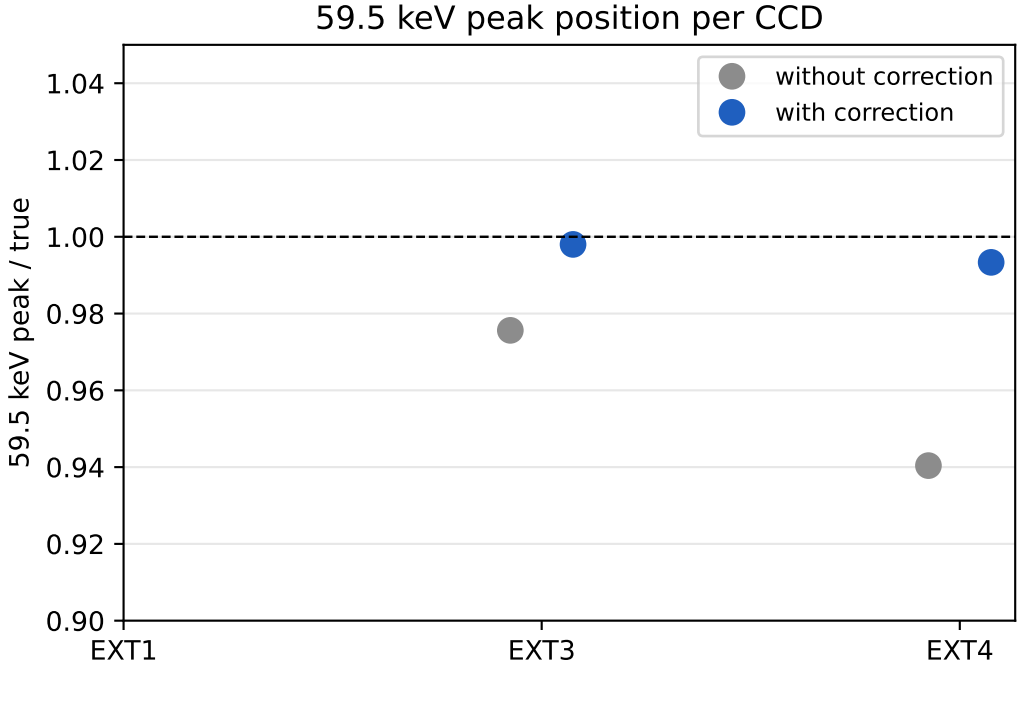
Reading it: 1.000 = right energy; '-' = no clear peak in that CCD (too few counts, or blended with the background of these partly flooded CCDs). With the correction the lines move towards 1.000 and the three CCDs agree better. What is left (0.2-0.7% low on the 59.5 keV line) is most likely charge in the faint edge of each cluster that panaSKImg's clustering threshold drops - not a gain effect, so the correction cannot recover it. Pages 2-3 show the spectra.

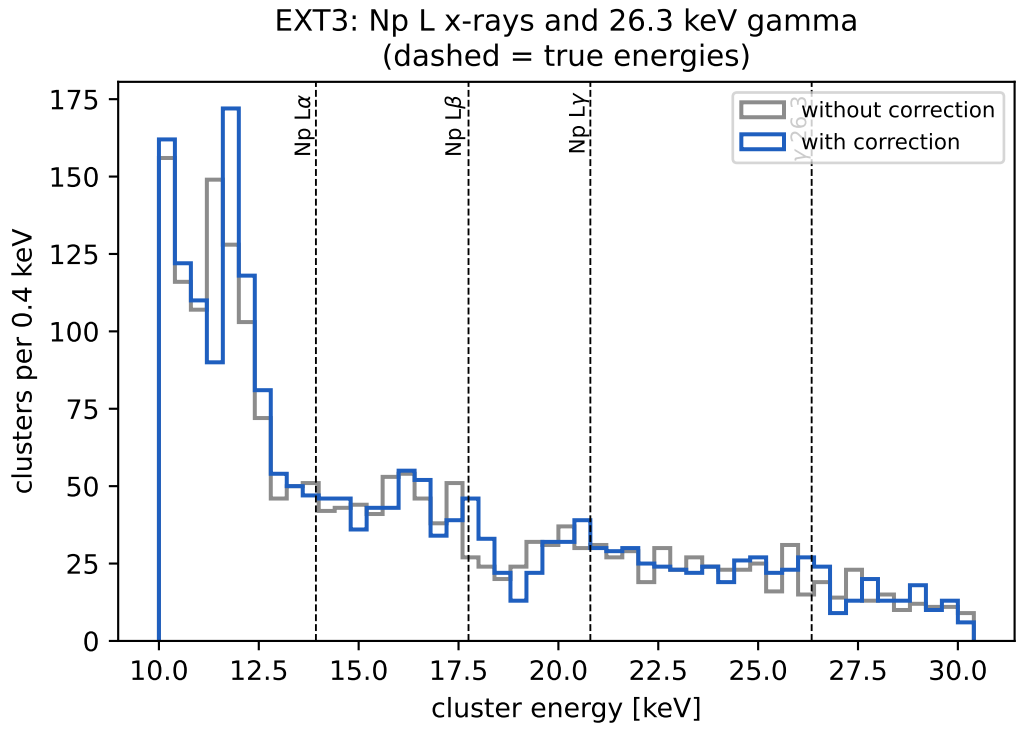
Page 2 - the 59.54 keV gamma line (and its Si escape peak 1.74 keV below)

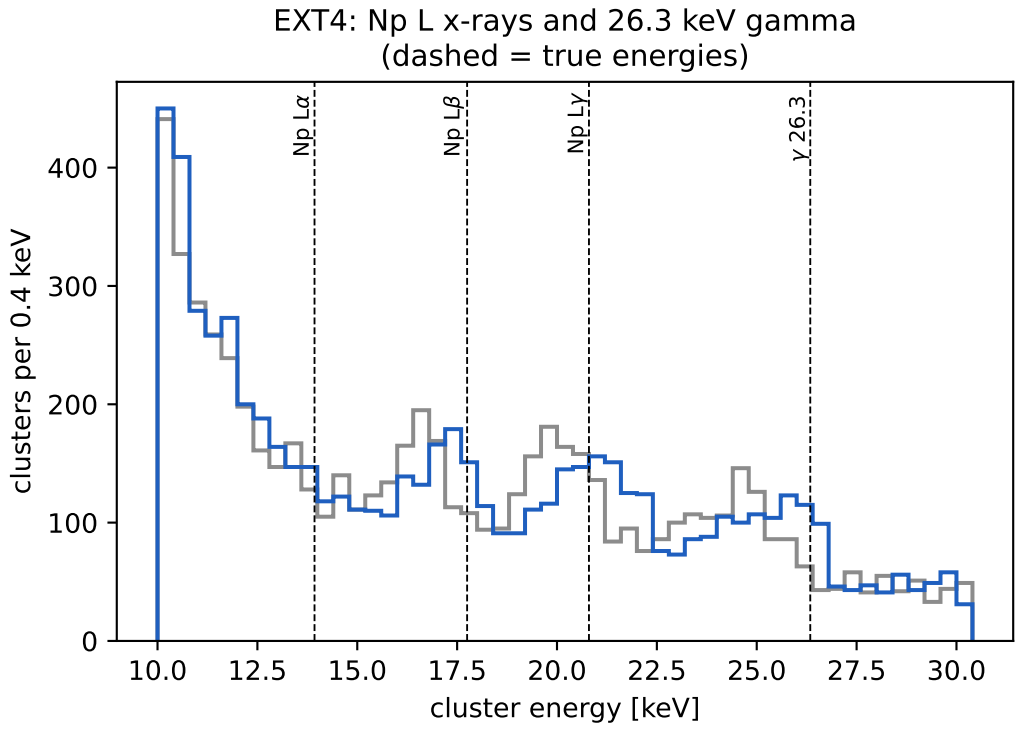


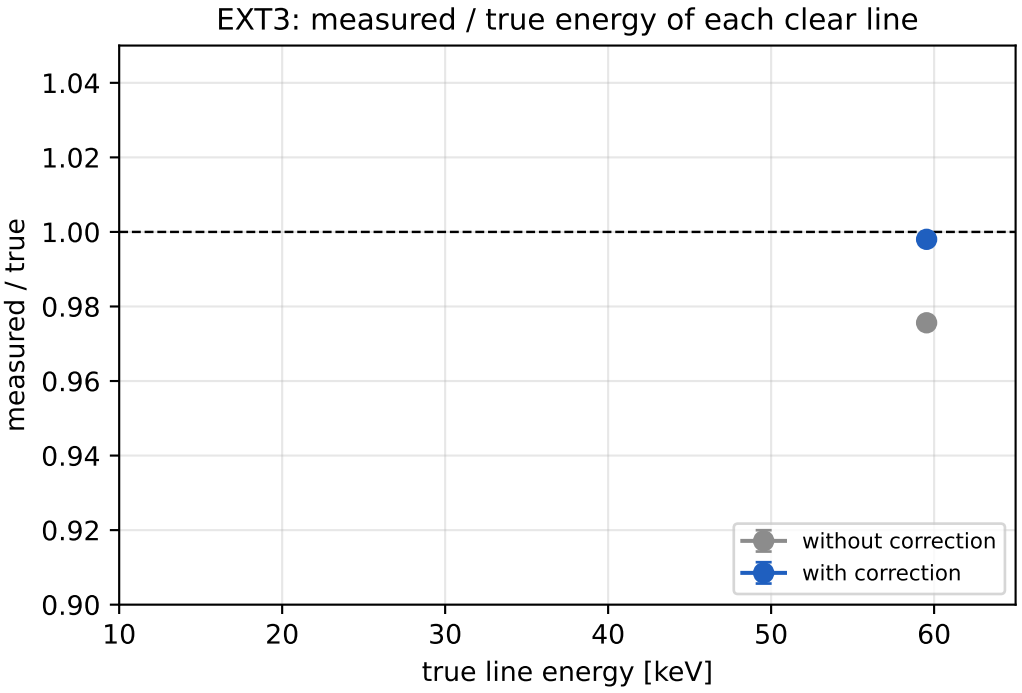


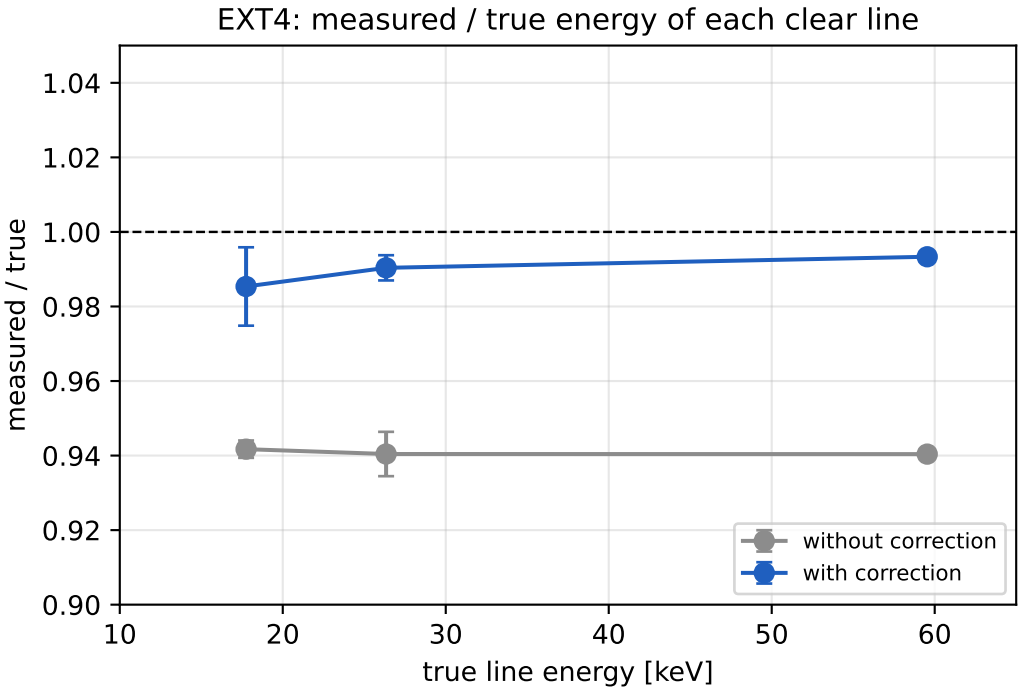





EXT3: Np L x-rays and 26.3 keV gamma
(dashed = true energies)
clusters per 0.4 keV
cluster energy [keV]
Np Lα
Np Lβ
Np Lγ
γ 26.3
without correction
with correction


EXT4: Np L x-rays and 26.3 keV gamma
(dashed = true energies)
clusters per 0.4 keV
cluster energy [keV]
Np Lα
Np Lβ
Np Lγ
γ 26.3


EXT3: measured / true energy of each clear line
measured / true
true line energy [keV]
without correction
with correction


EXT4: measured / true energy of each clear line
measured / true
true line energy [keV]
without correction
with correction